COMPTIA N+

N+ NETWORK FUNDAMENTALS AND BUILDING NETWORKS

CONFIGURE IPv6 ADDRESSING

TASKS:1

Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

RESULT

- Opened **Command Prompt** as Administrator.
- Checked the available network interfaces using: netsh interface ipv6 show interfaces
- Assigned a static IPv6 address to the Wi-Fi adapter using: netsh interface ipv6 add address "Wi-Fi" 2001:db8:1::10
- Verified the configuration by running: ipconfig
- Confirmed that the new IPv6 address was displayed under the Wi-Fi adapter.

TASKS:2

Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address. Use the ping6 or ping -6 command for IPv6 testing.

RESULT

1. Opened **Command Prompt** as Administrator.
2. Assigned a Unique Local IPv6 address starting with **fd**: netsh interface ipv6 add address "Wi-Fi" fd00:1::10

3. Verified the address by using: ipconfig
4. Tested the connection with another device on the same network: ping -6 fd00:1::20
5. Checked that the device responded successfully.

TASKS:3

Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.Document the steps and the observed address behavior in a text file.

RESULT

1. Opened **Command Prompt**.
2. Typed: **ipconfig**
3. Noted the Temporary IPv6 Address displayed.
4. Restarted the computer (or disconnected and reconnected the network).
5. Ran the ipconfig command again.
6. Compared the new Temporary IPv6 address with the previous one.
7. Recorded the observations in a text file.

TASKS:4

Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

RESULT

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#PYTHON

```
# Zomato-style Restaurant IPv6 Search


restaurants = [
    {"name": "Zomato Restaurant", "ipv6": "2001:db8:1::10"},
    {"name": "Dominos Pizza", "ipv6": "2001:db8:1::20"},
    {"name": "Pizza Hut", "ipv6": "2001:db8:1::30"},
    {"name": "Burger King", "ipv6": "2001:db8:1::40"},
    {"name": "Subway", "ipv6": "2001:db8:1::50"},
    {"name": "Cafe Coffee Day", "ipv6": "2001:db8:1::60"}
]


search = input("Enter restaurant name: ").lower()


found = False


for restaurant in restaurants:
    if search in restaurant["name"].lower():
        print("Restaurant:", restaurant["name"])
        print("IPv6 Address:", restaurant["ipv6"])
        found = True
```

```
if not found:
    print("No matching restaurant found.")
```

TAKE A EXMAPLE

Enter restaurant name: pizza

Restaurant: Dominos Pizza

IPv6 Address: 2001:db8:1::20

Restaurant: Pizza Hut

IPv6 Address: 2001:db8:1::30

RESULT: the program successfully searches restaurant using a partial name and display their IPv6 addresses

TASKS:5

Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

RESULT

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AI-Generated IPv6 Subnetting Scenario

A company has been assigned the IPv6 network: 2001:db8:1000::/48

The company creates three separate subnets for different departments:

Department	IPv6 Subnet
Sales	2001:db8:1000:1::/64
HR	2001:db8:1000:2::/64
IT	2001:db8:1000:3::/64

IPv6 Address Assigned

I selected the **Sales** subnet and assigned the following IPv6 address to my system: 2001:db8:1000:1::10/64

STEPS:-

1. Opened **Command Prompt** as Administrator.
2. Entered the following command: netsh interface ipv6 add address "Wi-Fi" 2001:db8:1000:1::10
3. Verified the assigned address using: ipconfig
4. Confirmed that the new IPv6 address was displayed successfully.